DSA & Time Complexity Cheat Sheet

Prepo by Avneesh — Big-O Analysis & Common Patterns | Category: FORMULA SHEET

1. Sorting Algorithm Complexities

- Quick Sort: Best/Avg: O(N log N) | Worst: O(N^2) | Space: O(log N) auxiliary recursion
- Merge Sort: Best/Avg/Worst: O(N log N) | Space: O(N) | Stable sorting
- Heap Sort: Best/Avg/Worst: O(N log N) | Space: O(1) in-place
- Insertion Sort: Best: O(N) (nearly sorted) | Worst: O(N^2) | Space: O(1)

2. Core Data Structure Operations

- Array / Dynamic Array: Access O(1), Search O(N), Insert/Delete O(N)
- Hash Map / Hash Set: Average Access/Insert/Delete O(1) | Worst: O(N) hash collisions
- Binary Search Tree (BST): Balanced: O(log N) search/insert | Skewed: O(N)
- Priority Queue / Binary Heap: Find Min/Max O(1), Insert O(log N), Extract Min/Max O(log N)